Project

Supply Chain Analytics

Analyzing supply chain's inefficiencies - Just in Time Company

Agenda

01 Briefing Recap

- Business performance
- Customer and Product
- Inventory management

02 Business Intelligence Analysis

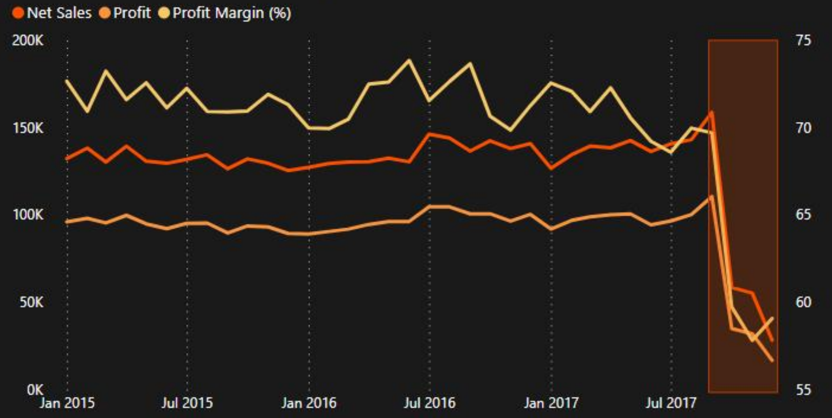
- Business downturn
- Overstock and understock
- Delay shipment

03 Proposal

01 Briefing Recap

1. Business performance

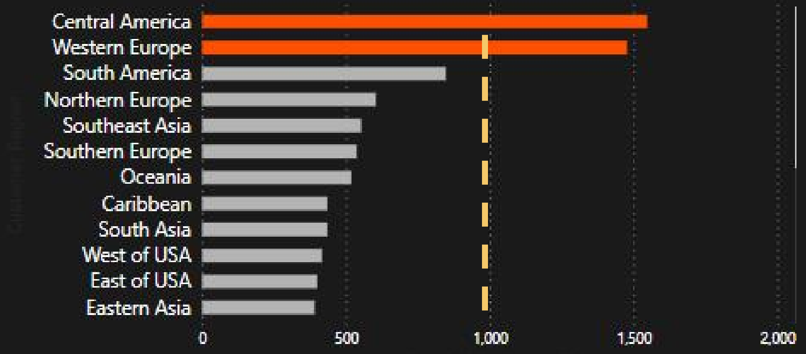
The company has had an impressive business performance over the past 3 years



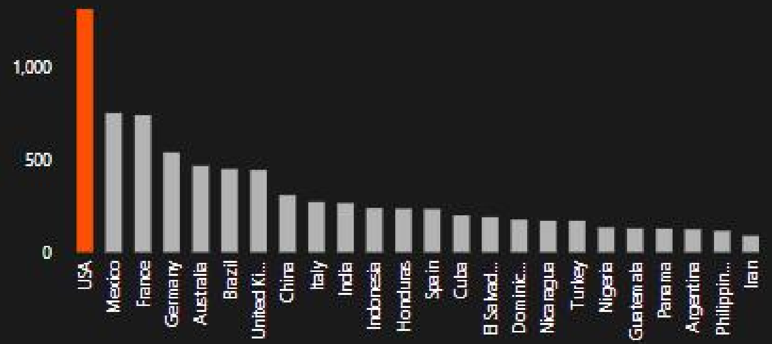
However, both net sales and profit **dropped sharply** in Q4/2017

2. Customer

The majority of customers come from Central America and Western Europe



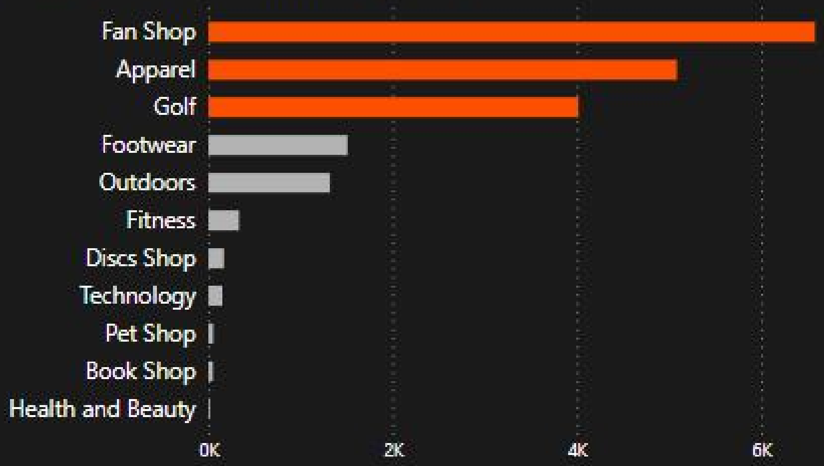
In which, customers from USA are much higher than other countries



3. Product

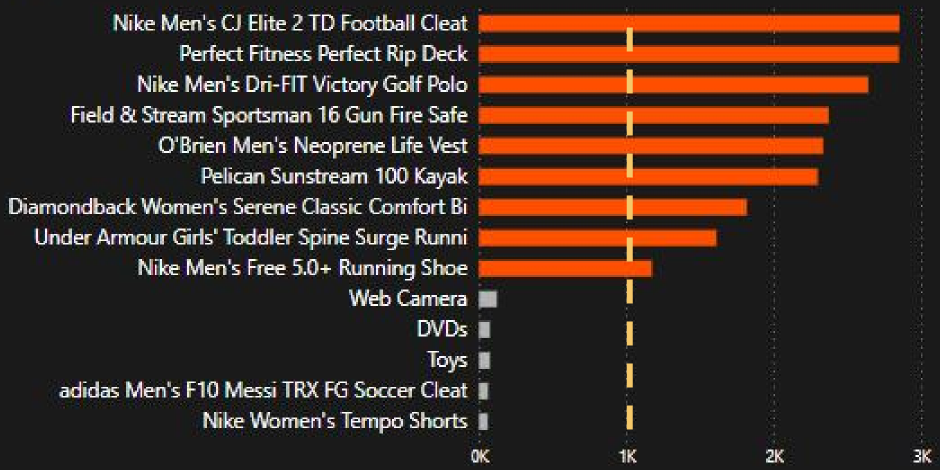
Apperal, Fanshop, and Golf are the most popular, outperforming the rest of the product departments

Product departments by Number of Orders



The gap between the top 9 product names in the lead and the product names lagged behind is considerable

Top 15 Product names by Number of Orders

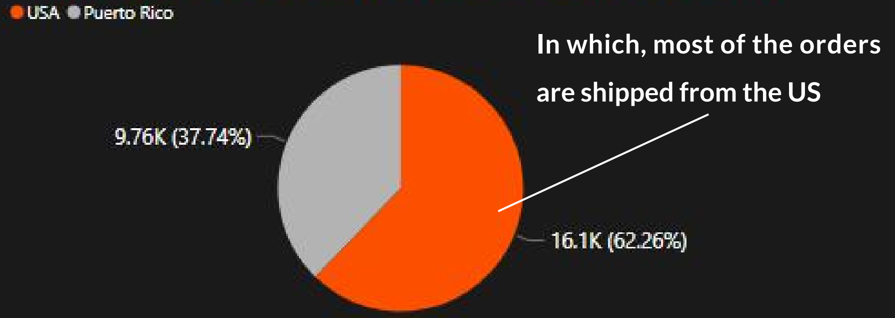


These three departments alone account for more than 70% of orders

There are no product names out of the top 9 could reach 1000 orders

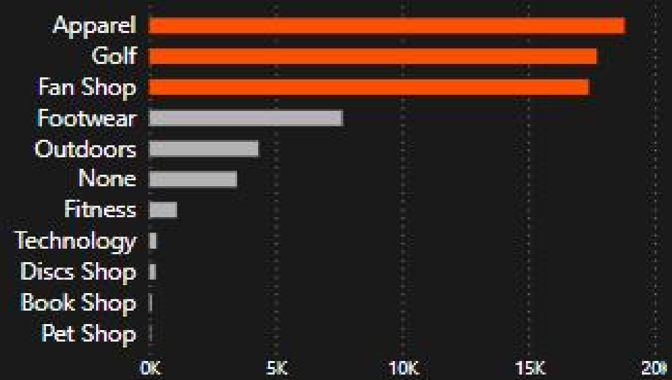
4. Inventory

The warehouses are only located in US and Peuter Rico



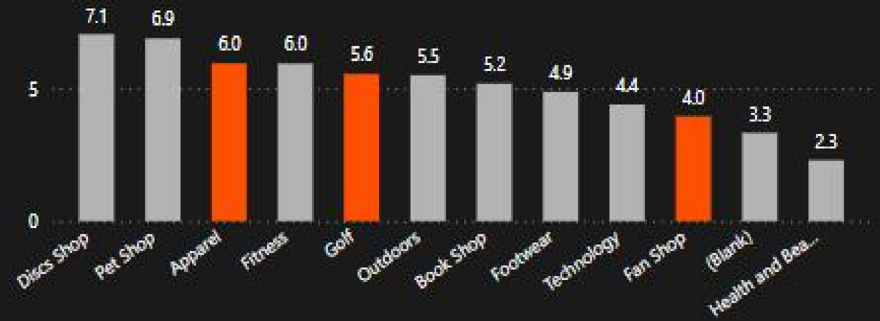
The best selling departments are also the ones with the greatest warehouse inventory

Warehouse Inventory by Product Department



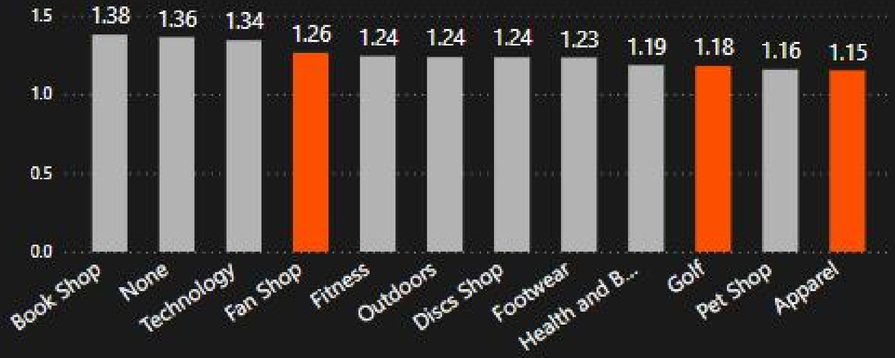
The time of fulfillment for the best seller department is quite long compared to the others, except for Fan Shop which can replenish quickly

Average of Warehouse Order Fulfillment (days) by Product Department



The opposite trend appeared in inventory price per unit with Fan Shop having higher average storage cost than Golf and Apparel

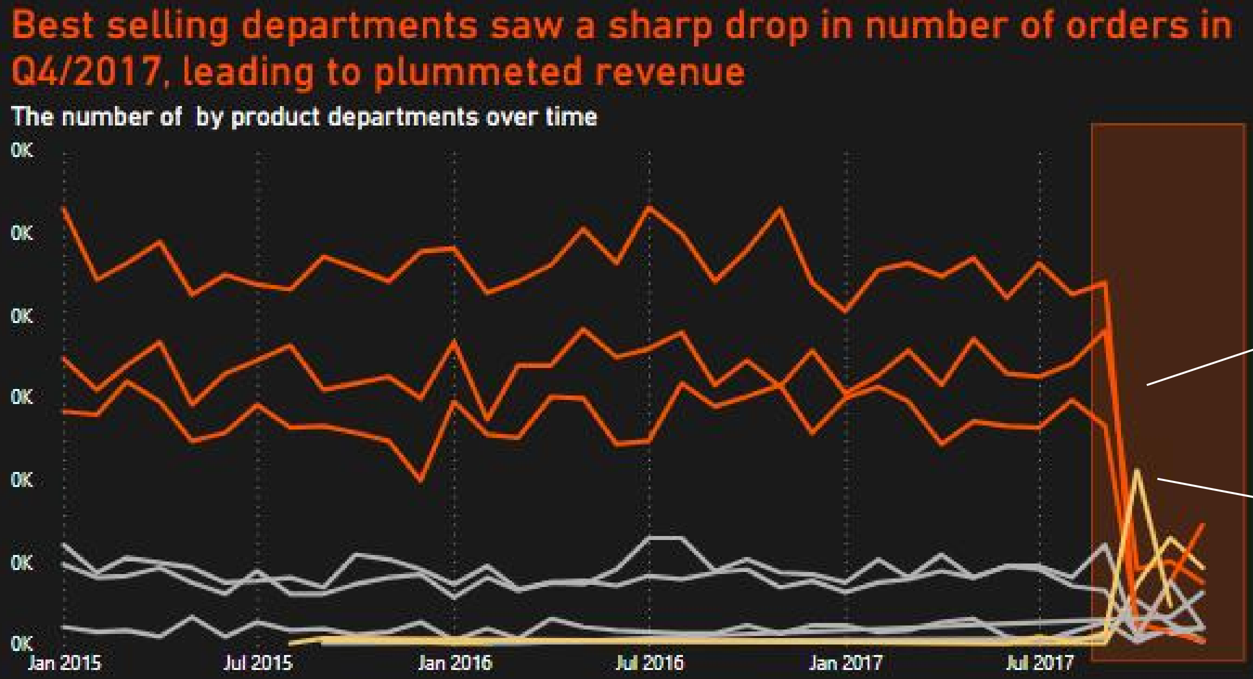
Average inventory cost per unit by Product department



02

Business Intelligence Analysis

1. Business downturn

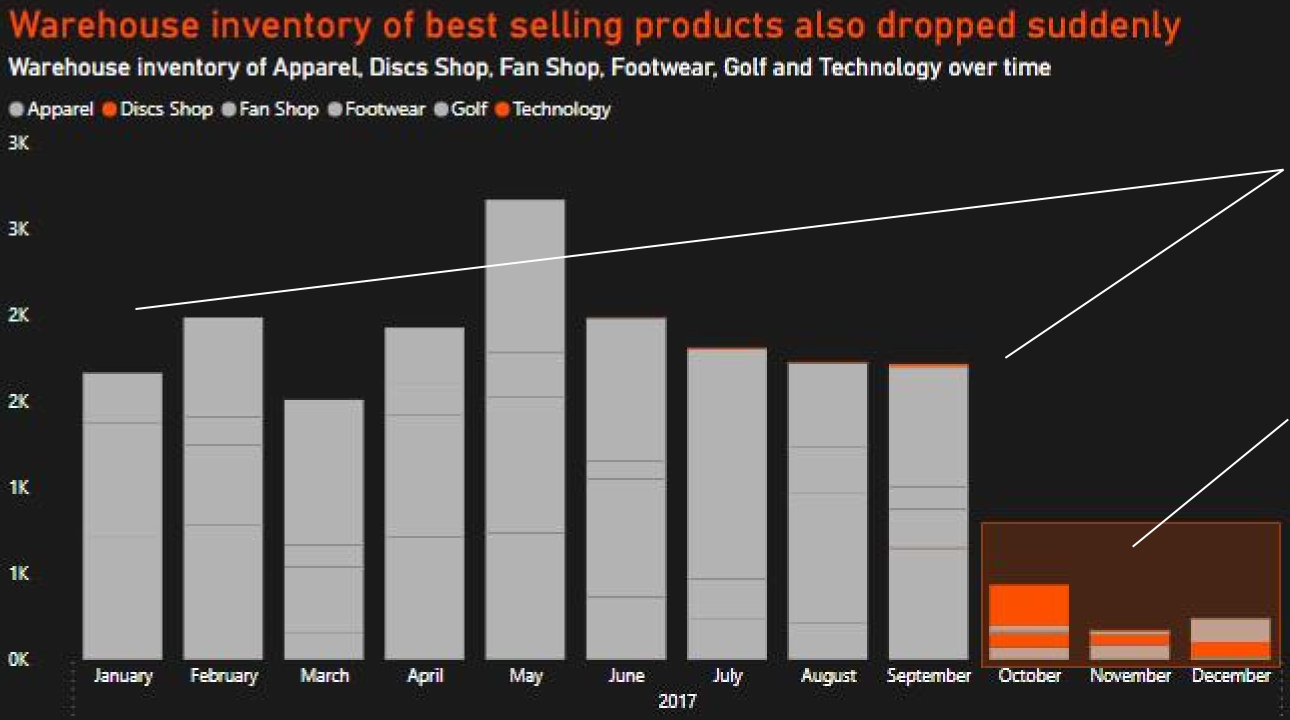


Apparel, Fanshop, Footwear and Golf which were instrumental to business's success has suddenly plummeted and almost disappeared.

Meanwhile, the product departments such as **Technology and Discs Shop** showed potential growth

The absence of the old leading - best selling departments (Apparel, Fanshop, Footwear, Golf), was closely linked to this downturn.

1. Business downturn

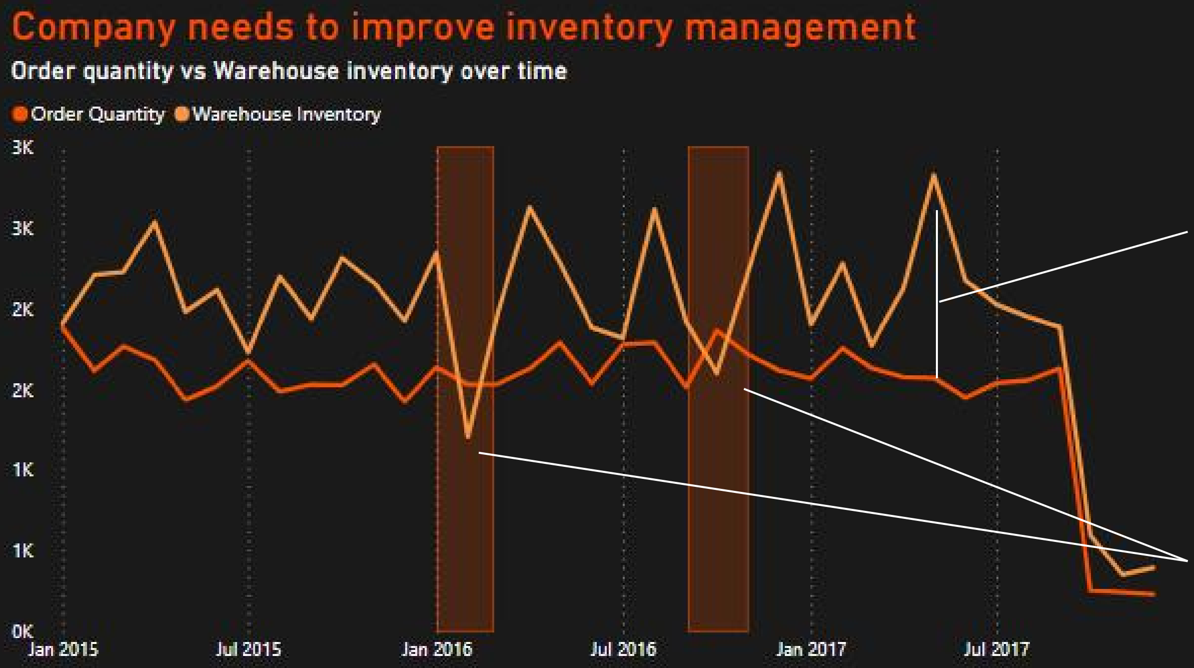


Inventories of <u>Apparel, Fanshop, Footwear and Golf</u> almost disappeared even though there was no sign of a sharp decrease before.

New departments like <u>Technology</u> suddenly had an <u>increase</u> in inventory follow by a gradual decrease

This decline was not due to changing customer preferences, but rather a supply chain disruption from supplier network

2. Inventory management



The company is frequently having excessive inventory or overstocked

Inventory fluctuates strongly at times, even though customers' demand is quite stable

The company is facing the challenge of not being able to maintain a stable warehouse inventory to meet the ordering demand of customers

3. Delay shipment

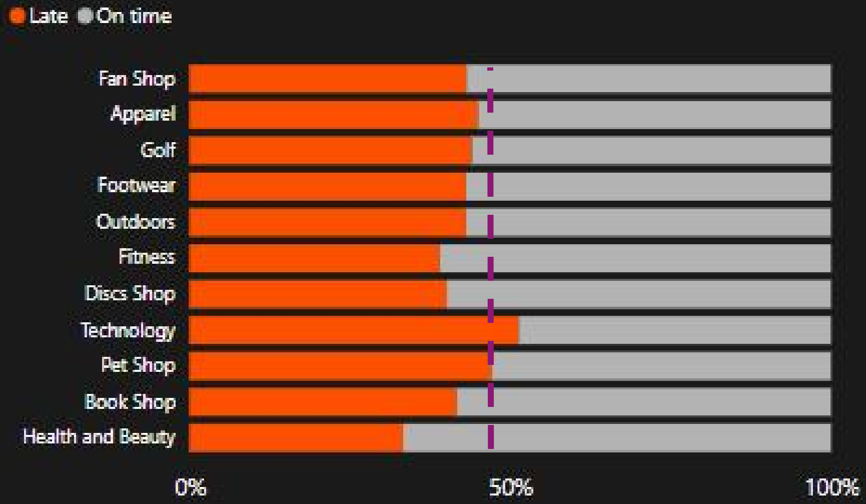
The company has a concerning late shipment rate

Late Shipment Rate over time (%)

LSR range from 40% to 50%

This high rate of late delivery occurs in all product departments

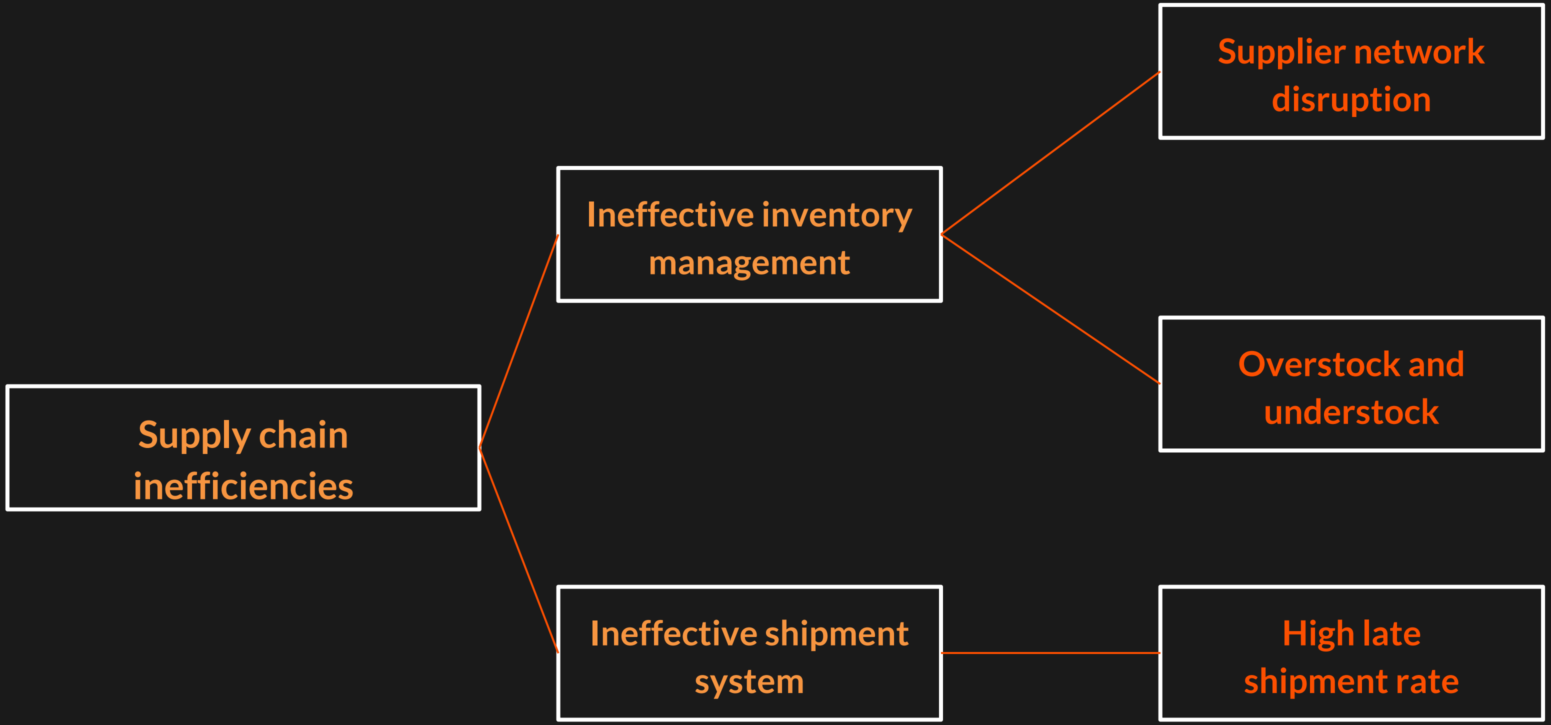
Percentage of Late Shipment and On time Shipment



Most product departments have LSRs around the 40% mark, except for technology whose rates go up to more than 50%.

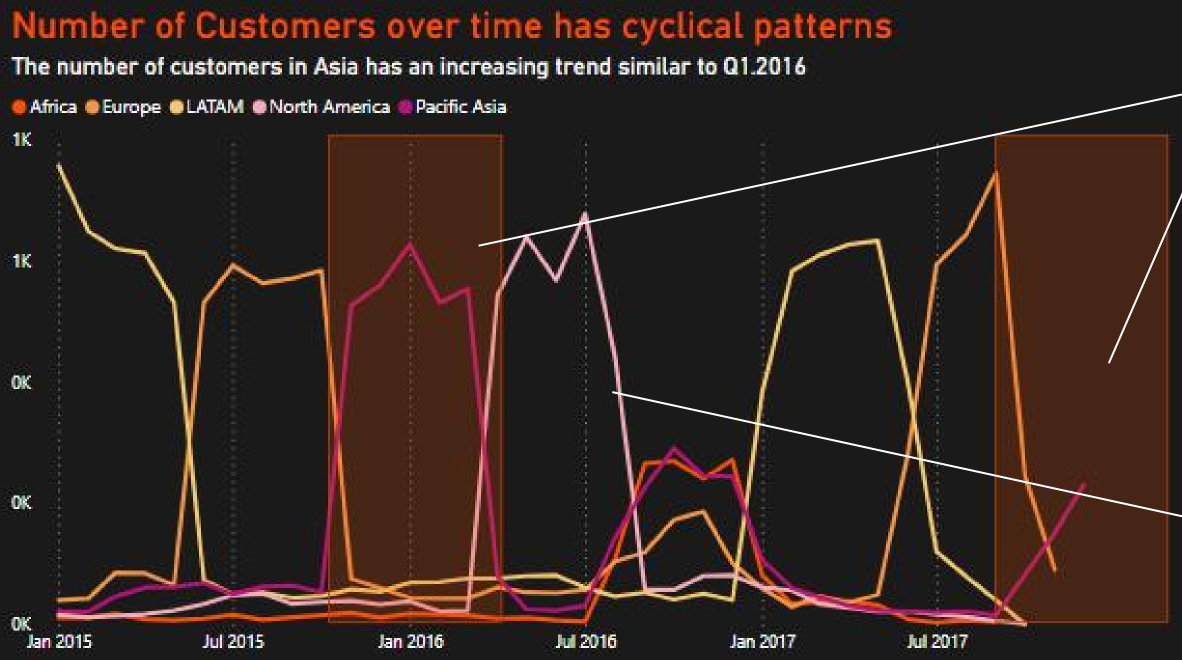
The high late shipment rate of average 40% over a long time which remains concerning and indicates inefficiencies in the shipping system that need improvement

The company is facing three main supply chain issues that need to be addressed



03 Proposal

1. Supplier disruption



In short term, the company should focus on suppliers in the likely growing customer group in Asia, especially those that provide emerging product departments such as technology.

In the long term, the next main customer group is likely to be from North America, the company needs to take advantage of the time to redesign the supplier network to recalling old best selling departments

Over time, the tendency to join shopping online rise sharply in the LATAM, then in turn to Europe, Pacific Asia, North America and Africa. This trend continues to repeat from the LATAM

2. Inventory management

Call to action		Objective
Forecasting demand	- Analyzing historical sales data - Tracking market trends - Employing predictive analytics	Anticipate future demand, allowing to maintain an optimal inventory level.
Setting up controlling indicators for each product	- Reorder points - Lead time - Sales velocity - Desired safety stock levels	Ensuring the company replenishes stock in a timely manner without risking overstocking
Predicting the product life cycle for each region (if possible)		Calculate the appropriate amount of inventory and chose the right product for each market

3. Delay shipment

Call to action		Objective
The markets near the warehouse (North America, LATAM, Western Europe,...)	- Redesigned transportation route - Adopting a cross-docking strategy - Take advantage of Less - than - truckload transportation	- Reduce the turnaround time - Reduce shipping time
The markets far away from the warehouse (Asia Pacific, SEA,...)	- Establishing a warehouse in a strategic logistics hub like Singapore in Asia	- Expand the company's control on global reach - Reduce the turnaround time - Reduce the shipping time to those markets
Both	- Collaboration with local logistics companies	- Leverage their local existing resources and expertise to optimize cost and supply chain

Thank you